

Application description

SignLingo is a full-stack web app where learners study American Sign Language (ASL) through various lessons, timed quizzes with levels, and live webcam practice powered by a lightweight ML detector. Track XP, and levels, and compete on the leaderboard—all with a clean, React-based UI and a Node/Express/MongoDB backend.

Features

- Auth: Signup/Login with cookie session; “Remember me” UI.
- Lessons: Browse a lesson grid, open details, and mark signs as done and practice live; progress bar + Reset Progress.
- Characters: Searchable ASL sign gallery (labels, categories, tags, images).
- Quizzes: Level-based quizzes with 40s timer, stars, per-level score, and total score get 3 questions right and unlock a level - 10 levels.
- Live Practice: Webcam capture → server endpoint for letters/gestures prediction; shows label + confidence.
- Profile: Edit profile, delete account, XP bar, level, lessons in progress, quiz stats.
- Leaderboard: Top 10 users based on XP displayed
- Right Rail: Daily goal progress and quick stats (XP / Level).
- Nav bar: Navigate through the app
- Auth routes: POST /auth/signup, POST /auth/login, POST /auth/logout, GET /auth/check (Express-session).
- Profile overview: single endpoint aggregates user stats, lesson progress, and quiz status.
- Lessons API: fetch lessons, start/continue a user-lesson, mark sign done, reset lesson progress.
- Quizzes API: get unlock status, start/check/finish quiz sessions, persist stars & total score.
- Live detect: proxy to a FastAPI service (/api/live/detect?mode=letters|gestures).

Users – Target audience (user personas)

Primary target users:

- **Beginner learners** who want to learn basic sign language (students, professionals, family members of Deaf/HoH people).
- **Hearing users** with little to no prior experience in sign language who are comfortable using web apps and webcams.

User Personas :

1. Kary – Undergraduate Student

Wants a structured way to learn signs to communicate on campus.

2. Gabriel – Community Volunteer

Needs fast, functional sign learning to support accessible events.

3. Meera – Parent of a Hard of Hearing Child

Wants repeated lessons and progress tracking to assist at home.

User Stories

- a) As a new learner, I want to browse signs by category to start my learning intuitively.
- b) As a registered user, I want to start lessons and view my progress history.
- c) As a busy learner, I want short, 10-question practice quizzes with immediate feedback.
- d) As a review-focused user, I want to revisit signs I got wrong or haven't practiced in a while.
- e) As a stats-driven learner, I want to see visual feedback (streaks, levels) on my dashboard.
- f) As an advanced learner, I want to try real-time sign-to-text feedback via webcam.
- g) As a caregiver or teacher, I want to reset lessons or view past quizzes for shared learning.

Data description

SignLingo stores and displays:

- User account data (name, email, hashed password)
- Profile information (XP, level, streaks, badges)
- Lesson content (titles, descriptions, videos/images, difficulty level)
- Quiz questions, answers, and scores
- Progress data (completed lessons, history of attempts, time spent)

Main tasks – Use cases

Core tasks for the study:

- **T0 – Initial approach (intuitiveness)**
Without any instructions, explore the homepage and tell us what you think this app does and how you would start learning.
- **T1 – Create an account and reach dashboard**
From the landing page, sign up as a new learner and reach your main learning dashboard (or log in if you already created an account earlier in the session).
- **T2 – Start and complete a lesson**
From the dashboard, find a beginner lesson (e.g., “ASL Alphabet – Level 1”), start it, complete the lesson content.
- **T3 –Start a Quiz**
From the dashboard, find a beginner lesson (e.g., “ASL Alphabet – Level 1”), start it, complete the lesson content.
- **T4 - Browse through Different character and check leaderboard**

Try to look up a character through the application and checkout the leaderboard scores
- **T5- Go to profile see what’s your progress**

In the profile page , Identify the different Progress you have made throughout the app
- **T6 - Try to Update your profile**

Try to update the profile details like name , email or age

Experiment – Steps

Welcome participants

Ask for consent to record

Ask to think out loud

Remind them they aren’t being evaluated

Ask participant to think like the target audience

Demographics questions (pre-test)

1. **Name**
 2. **Age**
 3. **Country/Region you currently live in:**
 4. **How often do you use web or mobile apps to learn something new (like Duolingo, YouTube tutorials, etc.)?**
 5. **How familiar are you with sign language?**
 6. **How comfortable are you using your computer's webcam in apps?**
 7. (Optional) **Do you have any visual, hearing, or motor conditions that might affect how you use this app?**
-

Post-test questionnaire – Likert scales

Use a **5-point Likert scale**. For effectiveness:

1 = Strongly disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly agree

For ease:

1 = Very difficult, 2 = Difficult, 3 = Neutral, 4 = Easy, 5 = Very easy

You'll have **3 participants**, each filling this once.

Task-specific questions

For Task 1 (create account + reach dashboard):

1. *Effectiveness – T1*
“The application allowed me to successfully create an account/log in and reach the main dashboard.”
Scale: 1–5 (Strongly disagree → Strongly agree)
2. *Ease of use – T1*
“Completing Task 1 (creating an account and reaching the dashboard) was...”
Scale: 1–5 (Very difficult → Very easy)

For Task 2 (complete lesson):

3. *Effectiveness – T2*

“I was able to find, start, and complete a lesson and its quiz using the application.”

Scale: 1–5

4. *Ease of use – T2*

“Completing Task 2 (lesson + quiz) was...”

Scale: 1–5

For Task 3 (Complete Quiz):

5. *Effectiveness – T3*

“The application successfully helped me practice using the Quiz .”

Scale: 1–5

6. *Ease of use – T3*

“Using the Quiz was Easy ”

Scale: 1–5

For Task 4 (Characters and leaderboard):

7. *Effectiveness – T3*

Scale: 1–5

8. *Ease of use – T3*

For Task 5 (Profile progress check):

9. *Effectiveness – T3*

10. *Ease of use*

For Task 6 (Update Profile):

11. *Effectiveness*

12. *Ease of use*

Overall questions

7. *Overall effectiveness*

“Overall, SignLingo was effective for helping me learn and practice sign language.”

Scale: 1–5

8. *Overall ease of use*

“Overall, SignLingo was intuitive and easy to use.”

Scale: 1–5

9. *Likelihood of use*

“I would like to use an app like SignLingo in real life to learn sign language.”

Scale: 1–5 (Strongly disagree → Strongly agree)

Final open question

10. **Any final comments or suggestions for improvement?**

Participant sections

Participant 1

Demographics :

1. Name : ***Thej Kiran***
2. Age : **22**
3. Country/Region you currently live in: ***United Kingdom***
4. How often do you use web or mobile apps to learn something new (like Duolingo, YouTube tutorials, etc.)? : ***Often***
5. How familiar are you with sign language? : ***No***
6. How comfortable are you using your computer's webcam in apps? : ***Comfortable***
7. (Optional) Do you have any visual, hearing, or motor conditions that might affect how you use this app? : ***No***

Tasks :

T0 – Initial approach

- Completed? Yes / No - **YES**

- Issues/confusions: **Start learning button did not work**

- Notable quotes/observations:

- User was unable to use the start learning button
- User identified the purpose of the app pretty correctly
- User was able to identify the login button and successfully go to the signup login form

T1 – Create account + dashboard

- Completed? Yes / No - **YES**

- Errors or help needed: Could not login using Google account

- Observations:

- If login through google account worked, it would have saved time for the user
- Otherwise creating an account was pretty straightforward

T2 – Complete lesson

- Completed? Yes / No : **Yes**

- Errors or help needed: User was confused about the missing images

- Observations:

- Missing images are because of time constraint of the project timeline , we could not generate all the images to put in the App
- User was able to understand and learn from the lesson

T3 – Complete quiz

- Completed? Yes / No : **YES**

- Errors or help needed: User was confused about the missing images , There was no success message after the answer was correct so user got confused

- Observations:

- should implement a success message

T4 – Characters + leaderboard

- Completed? Yes / No : **yes**

- Errors or help needed: None

- Observations: User was able to find and look through the leaderboard and characters page

T5 – Profile progress check

- Completed? Yes / No : **YES**
- Errors or help needed: User was able to identify everything
- Observations:
 - User was able to understand the whole profile page

T6 – Update profile

- Completed? Yes / No : **YES**
- Errors or help needed: User wanted to update profile picture , otherwise everything was clear
- Observations:
 - An option to update profile picture would have been a good addition

Post-test ratings:

- T1 Effectiveness: **5**
- T1 Ease: **5**
- T2 Effectiveness: **4.5**
- T2 Ease: **4.5**
- T3 Effectiveness: **5**
- T3 Ease: **5**
- T4 Effectiveness: **4**
- T4 Ease: **5**
- T5 Effectiveness: **4.5**
- T5 Ease: **4.5**
- T6 Effectiveness: **5**
- T6 Ease: **5**
- Overall effectiveness: **4.5**

- Overall ease of use: **4.5**

Key issues & takeaways (Participant 1)

- Suggestions / comments (from participant):

Adding a payment option, making it a paid app

An option to change language

Participant 2

Demographics :

1. Name : ***Parvathy Dinakar***
2. Age : **23**
3. Country/Region you currently live in: ***United Kingdom***
4. How often do you use web or mobile apps to learn something new (like Duolingo, YouTube tutorials, etc.)? : ***Often***
5. How familiar are you with sign language? : ***No***
6. How comfortable are you using your computer's webcam in apps? : ***Comfortable***
7. (Optional) Do you have any visual, hearing, or motor conditions that might affect how you use this app? : **No**

T0 – Initial approach

- Completed? Yes / No **YES**

- Issues/confusions: **No issues**

- Notable quotes/observations: User was able to identify the purpose of the app and was able to navigate accurately

T1 – Create account + dashboard

- Completed? Yes / No **YES**

- Errors or help needed: Google account did not work

- Observations: Could have been better if google account was implemented

T2 – Complete lesson

- Completed? Yes / No **YES**
- Errors or help needed: No help needed , User was able to navigate through the lessons
- Observations: Live practice could have been deployed correctly

T3 – Complete quiz

- Completed? Yes / No : **YES**
- Errors or help needed: Images confused the user
- Observations: Generation of all the images would have made the app more user friendly

T4 – Characters + leaderboard

- Completed? Yes / No : **YES**
- Errors or help needed: no help need , user was able to understand and navigate through the pages with ease
- Observations: Overall pretty user friendly UI

T5 – Profile progress check

- Completed? Yes / No **YES**
- Errors or help needed: Not much help needed , user was able to understand the profile page and what the purpose of it was.
- Observations: Overall it was user friendly

T6 – Update profile

- Completed? Yes / No : **YES**
- Errors or help needed: user wanted to update the profile picture , which was not enabled in the app

- Observations: profile picture could have been a good addition

Post-test ratings :

- T1 Effectiveness: **4**

- T1 Ease: **4**

- T2 Effectiveness: **4.5**

- T2 Ease: **4.5**

- T3 Effectiveness: **4**

- T3 Ease: **4**

- T4 Effectiveness: **4.5**

- T4 Ease: **4.5**

- T5 Effectiveness: **4**

- T5 Ease: **4**

- T6 Effectiveness: **4**

- T6 Ease: **4**

- Overall effectiveness: **4.5**

- Overall ease of use: **4.5**

- Suggestions / comments (from participant):

User wanted to know more about live practice sessions

Participant 3

Demographics :

1. Name : **Adithya**
2. Age : **25**
3. Country/Region you currently live in: **Boston , US**
4. How often do you use web or mobile apps to learn something new (like Duolingo, YouTube tutorials, etc.)? : **Often**
5. How familiar are you with sign language? : **No**

6. How comfortable are you using your computer's webcam in apps? : **Comfortable**
7. (Optional) Do you have any visual, hearing, or motor conditions that might affect how you use this app? : **No**

T0 – Initial approach

- Completed? Yes / No **YES**
- Issues/confusions: User was able to navigate through and understand the starting page

T1 – Create account + dashboard

- Completed? Yes / No **YES**
- Errors or help needed: User struggled to remember password and user wanted a forgot password option
- Observations: Implementing a forgot password option would make the app more user friendly

T2 – Complete lesson

- Completed? Yes / No **YES**
- Errors or help needed: User was able to understand the lessons clearly , User was little confused about the images
- Observations: overall user friendly

T3 – Complete quiz

- Completed? Yes / No **YES**
- Errors or help needed: User was unable to understand the quiz because of the lack of image availability but other than that pretty user friendly

T4 – Characters + leaderboard

- Completed? Yes / No **Yes**

- Errors or help needed: User struggled to identify the purpose of the characters page and leaderboard
- Observations: Maybe adding instructions about the page would make it more user friendly

T5 – Profile progress check

- Completed? Yes / No **YES**
- Errors or help needed: user was able to understand this page although the page went to not found when clicked on continue lesson or quiz and had to login again
- Observations: Fix the redirecting rewrite errors in deployment

T6 – Update profile

- Completed? Yes / No **YES**
- Errors or help needed: User was able to understand and update profile although delete profile led to the not found page
- Observations: Fix the redirecting rewrite errors in deployment

Post-test ratings

- T1 Effectiveness: **3**
- T1 Ease: **5**
- T2 Effectiveness: **4.5**
- T2 Ease: **4.5**
- T3 Effectiveness: **3.0**
- T3 Ease: **3.0**
- T4 Effectiveness: **3.0**
- T4 Ease: **3.0**
- T5 Effectiveness: **4.5**
- T5 Ease: **4.5**

- T6 Effectiveness: **3.0**
- T6 Ease: **3.0**
- Overall effectiveness: **3.0**
- Overall ease of use: **3.0**

Key issues & takeaways (Participant 3)

- Suggestions / comments (from participant): User wanted to implement forgot password option , integrate both characters page and lessons.